parameter_iteration

- Timestamp: `2026-04-21 05:20:31`
- Source file: `/ceph/project/sharmalab/dnimrich/cd8atlas/code/pipeline_elements.py`

Loading and preprocessing data

Loading from ../../data/subsampled_100000.h5ad

Loaded adata with with shape (100000, 19957)

Initialized `counts` layer from adata.X

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1. Quality Control

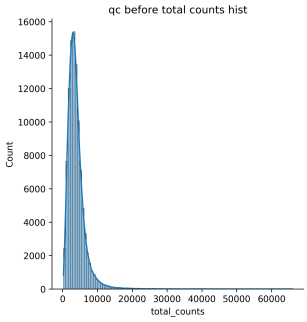
1.1 Ambient RNA QC

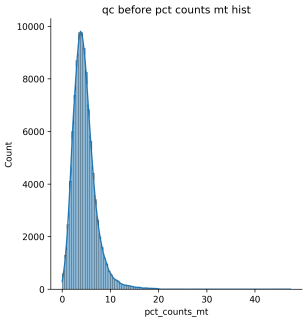
Checking if a cleanup is necessary, flagging cells with less than 1000 counts as potential ambient RNA contamination

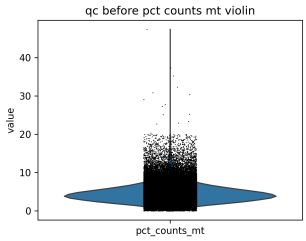
No significant RNA contamination detected, skipping cleanup

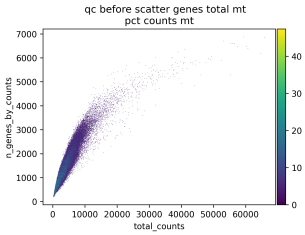
1.2 Count QC

index	count	mean	std	min	25%	50%	75%	max
total_counts	100000.00	3924.63	2698.54	248.00	2350.00	3424.00	4779.00	65975.00
n_genes_by_counts	100000.00	1434.59	638.66	201.00	1009.00	1346.00	1730.00	6872.00
pct_counts_mt	100000.00	4.61	2.41	0.00	3.02	4.26	5.73	47.37
pct_counts_ribo	100000.00	28.53	9.60	0.68	21.57	27.68	35.02	64.87
pct_counts_hb	100000.00	0.01	0.23	0.00	0.00	0.00	0.00	49.51
pct_counts_in_top_20_genes	100000.00	20.88	3.43	8.44	18.47	20.55	22.95	59.58









Number of count outliers with `MAD > 5`: 1548

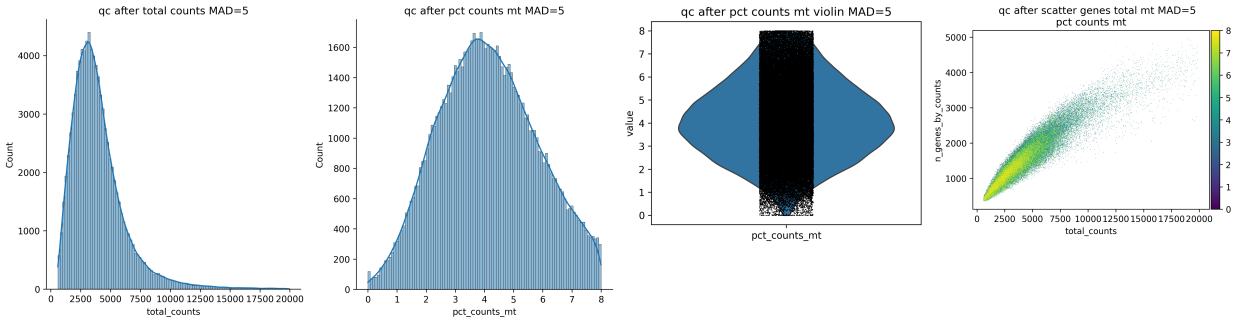
Number of mitochondrial outliers with `MAD > 5` or `pct_counts_mt > 8`: 7739

Filtered out outliers: 8902 (8.90%)

Filtering out genes expressed in less than 20 cells

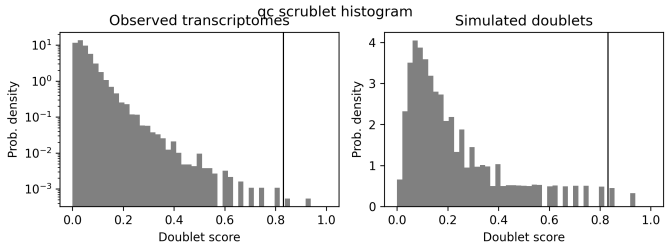
Removed 14368 genes (28.01%)

After count QC:



1.3 Doublet QC

Running scrublet with expected doublet rate 0.05



There are 2 predicted doublets, applying filter

After doublet step: (91096, 14368) (removed 0.00% cells)

1.4 Gene exclusion

Excluding genes matching the following patterns:

Note: Showing at most 10 rows and 10 columns from data with 11 rows and 2 columns.

mt_genes	['MT-']
ribo_genes	['RPS', 'RPL']
hb_genes	['^HBA|^HBB']
stress_genes	['G0S2', '^JUN', '^FOS', 'CDKN1A']
IFN_genes	['^IRF', '^IFN']
ccl_genes	['CCL']
MHC_genes	['^HLA-[A-Z]']
hist_genes	['HIST']

comp_genes	['^C1Q']
ig_genes	['^IGH', '^IGK', '^IGL']

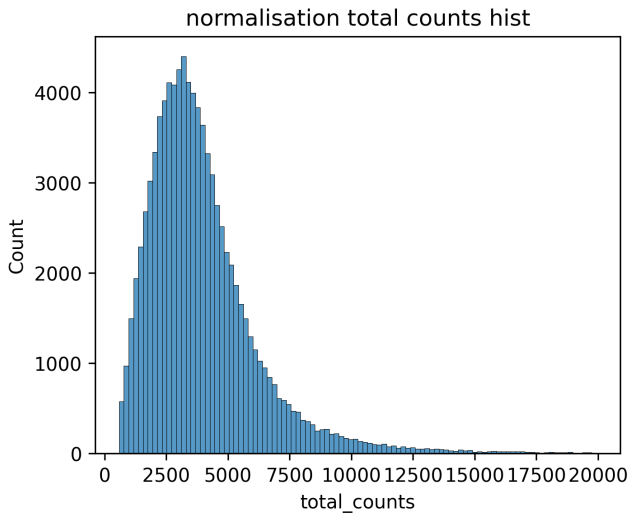
Removing 343 genes (2.39%)

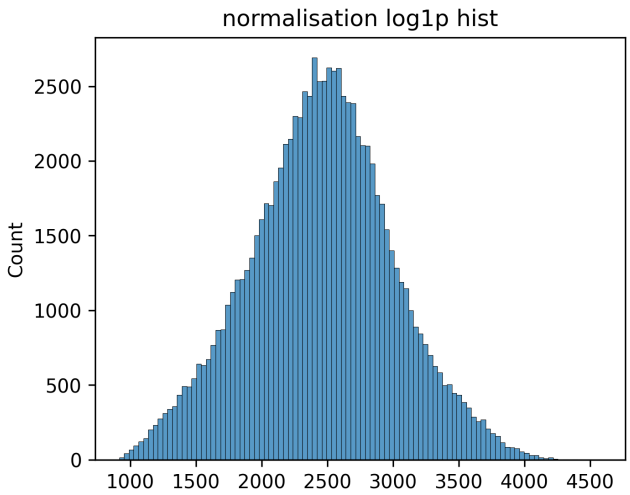
Shape after exclusion panel filtering: (91096, 14025)

2. Normalisation

Normalised in reference to total counts

Transformed with log1p





Hardcoded parameter grid

parameter	values
hvg_n_top_genes	[4000]
n_pcs	[25]
n_neighbors	[8, 10, 12]
min_dist	[0.1, 0.2]
leiden_resolution	[0.5, 0.5, 0.6]

Running 9 clustering evaluations (1 HVG settings x 1 PC settings x 3 neighbor settings x 3 Leiden resolutions)

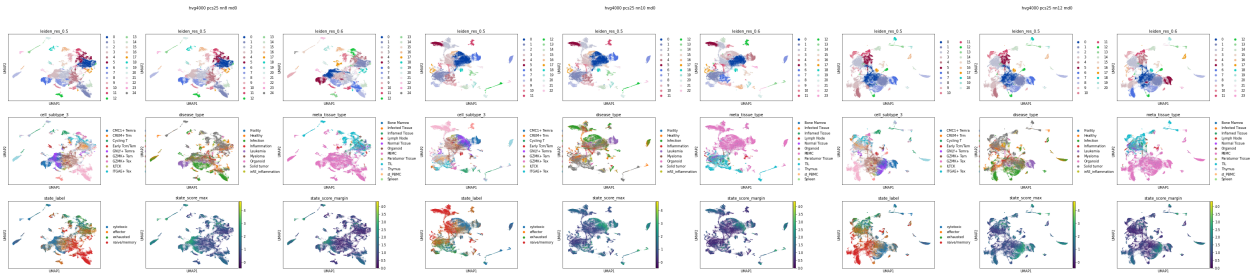
Rendering 6 UMAP panels for visualization (2 min_dist settings for each clustering graph)

UMAP summary grid

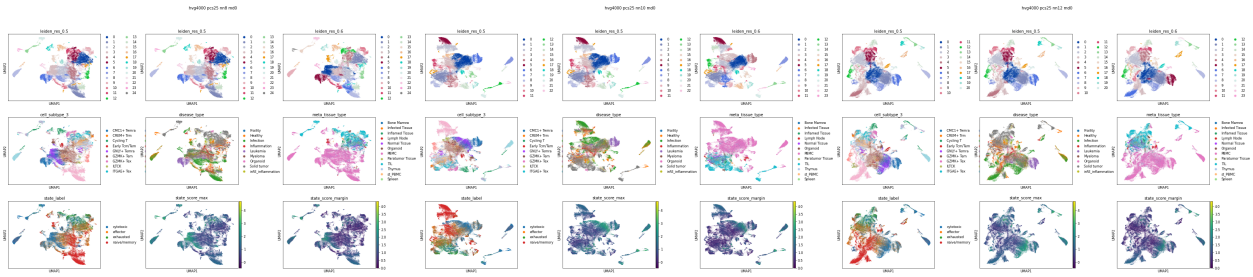
HVG 4000, PCs 25

Columns are n_neighbors = 8, 10, 12. Each panel contains all Leiden resolutions: 0.5, 0.5, 0.6.

min_dist = 0.1



min_dist = 0.2



Saving into parameter_iteration_20260421_052031_data/parameter_iteration_summary_statistics.csv

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Summary statistics

These statistics summarize clustering runs only. UMAP `min_dist` is treated as a visualization parameter and is not included in the CSV or summary plots.

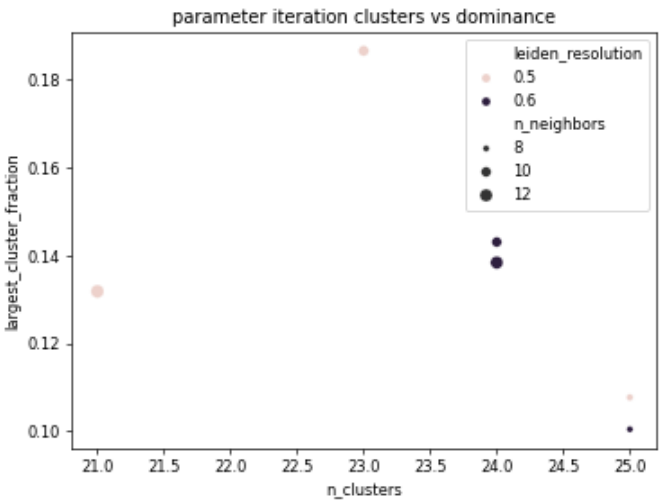
Full summary statistics CSV: parameter_iteration_20260421_052031_data/parameter_iteration_summary_statistics.csv

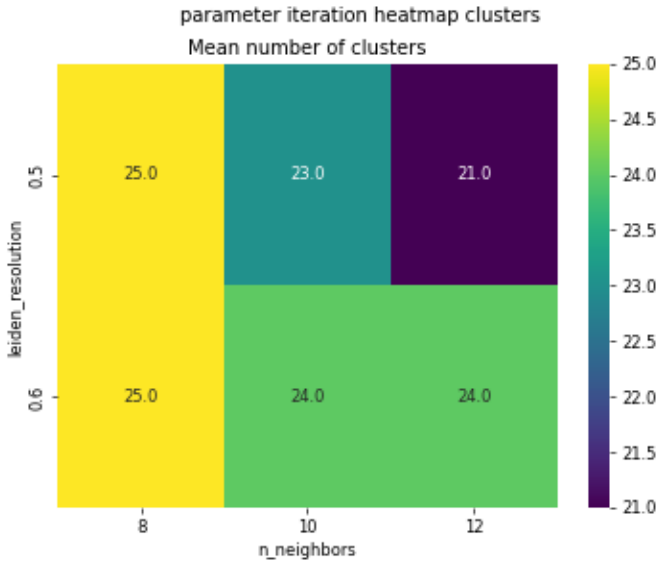
Note: Showing at most 50 rows and 10 columns from data with 9 rows and 12 columns.

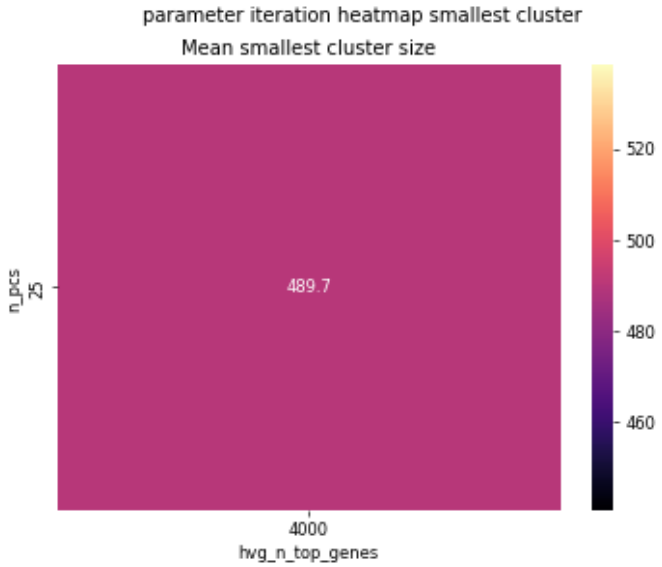
index	run_no	name	base_name	n_cells	n_genes	hvg_n_top_genes	n_pcs	n_neighbors	leiden_resolution	n_clusters
0.00	1.00	hvg4000_pcs25_nn8_res0.5	hvg4000_pcs25_nn8	91096.00	4000.00	4000.00	25.00	8.00	0.50	25.00
1.00	2.00	hvg4000_pcs25_nn8_res0.5	hvg4000_pcs25_nn8	91096.00	4000.00	4000.00	25.00	8.00	0.50	25.00
2.00	3.00	hvg4000_pcs25_nn8_res0.6	hvg4000_pcs25_nn8	91096.00	4000.00	4000.00	25.00	8.00	0.60	25.00

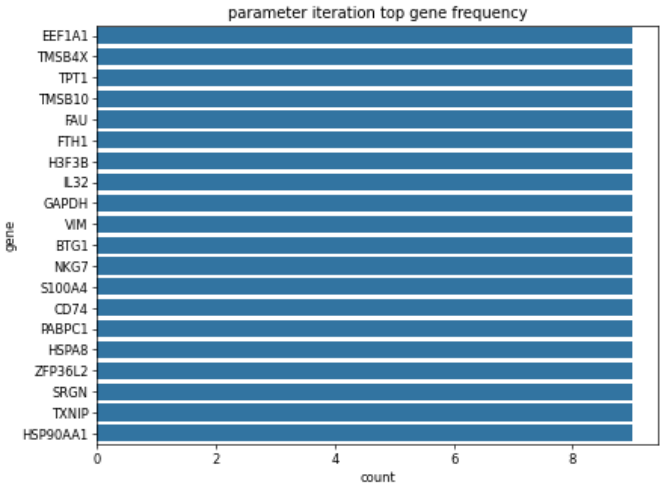
index	run_no	name	base_name	n_cells	n_genes	hvg_n_top_genes	n_pcs	n_neighbors	leiden_resolution	n_clusters
3.00	4.00	hvg4000_pcs25_nn10_res0.5	hvg4000_pcs25_nn10	91096.00	4000.00	4000.00	25.00	10.00	0.50	23.00
4.00	5.00	hvg4000_pcs25_nn10_res0.5	hvg4000_pcs25_nn10	91096.00	4000.00	4000.00	25.00	10.00	0.50	23.00
5.00	6.00	hvg4000_pcs25_nn10_res0.6	hvg4000_pcs25_nn10	91096.00	4000.00	4000.00	25.00	10.00	0.60	24.00
6.00	7.00	hvg4000_pcs25_nn12_res0.5	hvg4000_pcs25_nn12	91096.00	4000.00	4000.00	25.00	12.00	0.50	21.00
7.00	8.00	hvg4000_pcs25_nn12_res0.5	hvg4000_pcs25_nn12	91096.00	4000.00	4000.00	25.00	12.00	0.50	21.00
8.00	9.00	hvg4000_pcs25_nn12_res0.6	hvg4000_pcs25_nn12	91096.00	4000.00	4000.00	25.00	12.00	0.60	24.00

Parameter iteration summary plots




parameter iteration heatmap clusters
Mean number of clusters
25.0
23.0
21.0
25.0
24.0
24.0
leiden_resolution
0.5
0.6
8
10
12
n_neighbors
25.0
24.5
24.0
23.5
23.0
22.5
22.0
21.5
21.0


parameter iteration heatmap smallest cluster
Mean smallest cluster size
489.7
n_pcs
25
4000
hvg_n_top_genes
520
500
480
460


parameter iteration top gene frequency
EEF1A1
TMSB4X
TPT1
TMSB10
FAU
FTH1
H3F3B
IL32
GAPDH
VIM
BTG1
NKG7
S100A4
CD74
PABPC1
HSPA8
ZFP36L2
SRGN
TXNIP
HSP90AA1
gene
0
2
4
6
8
count